

1 Random Fields Supporting Quasi-Zero Modes in Nonperturbative
2 QED:
3 Connection with Asymptotic Freedom

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Abstract

It is found that any QED process calculated from a functional integral requires new, non-perturbative terms in its effective action. They follow from a rearrangement of Schwinger's proper time representation of the action.

The new terms are calculated from the bound states arising from random potential wells generated by the path integral. Their calculation is analogous to finding the bound states of a charged fermion in a four-dimensional random magnetic field-four dimensional because the calculation is in Euclidean space.

Such potentials are likely to contain broad, deep wells leading to closely spaced bound states. Every bound state contributes a negative term to the effective action, weighted by its degeneracy. The potentials generating these bound states are referred to as quasi-zero modes. Due to their probable large number, they can cause the effective action to turn negative from large amplitude-strong magnetic field-variations of the functional integral. The wells, being random, require no special symmetry. Such variations can cause the functional integral to converge.

8 I. INTRODUCTION

9 The question arises: can any QED process represented by a functional integral-path in-
10 tegral be calculated non-perturbatively?

11 As part of the electroweak model this question is of more than routine interest. The general
12 belief is that it cannot due to the integral's presumed unbounded, large-field growth subse-
13 quent to QED's lack of asymptotic freedom. [An exception occurs when all fermion loops
14 are neglected. But this is not QED.] Any attempt to elevate belief to certainty will require
15 non-perturbative information on QED's effective action.

16 One has every reason to ask what can possibly be new here. After all, we are starting from
17 work of Schwinger dating from the 1950s. The answer is that a huge amount of information
18 was overlooked in his 1951 paper [1] until years later. By simply rearranging Schwinger's
19 expression for the effective action, new non-perturbative information was finally unlocked
20 by mathematical theorems dating from the 1970s relating to constructive field theory [2–17].
21 The realization that some of them could be applied to four-dimensional QED came in 2015

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22 [18]. It took eleven years to apply this information to the specific questions above.

23 Consider, for example, the functional integral representation of the Euclidean Green func-
 24 tion for $2n$ external fermions and m photons

$$S_{\mu_1, \dots, \mu_m}(x_1, \dots, x_n : y_1, \dots, y_n : z_1, \dots, z_m) \\
= Z^{-1} \int d\mu_0(A) \det_{\text{ren}}(1 - \underline{e}S\mathcal{A}) \det[S(x_i, y_j | \underline{e}A)]_{i,j=1}^n \prod_{k=1}^m A_{\mu_k}(z_k), \quad (1)$$

25 where $S(x_i, x_j | A)$ is the fermion propagator in the external potential A_μ Z is the vacuum-
 26 to-vacuum amplitude that will eventually cancel out. The regularized fermion determinant
 27 $\det_{\text{ren}}$, written as $\exp(\ln \det_{\text{ren}})$, introduces the effective action $\ln \det_{\text{ren}}(1 - \underline{e}S\mathcal{A})$, where S
 28 is the free fermion propagator. It is a non-linear, non-local gauge invariant quantity. Its
 29 expansion in powers of $\underline{e}S\mathcal{A}$ generates the loop expansion which, when truncated, results in
 30 perturbation theory and is, ipso facto, unsuited to the above question. The subscript ren
 31 designates renormalization subtractions detailed below.

32 Further to [1], the potentials belong to a restricted class of random, tempered distributions
 33 belonging to $\mathcal{S}'(\mathbb{R}^4)$ that support the functional integral's Gaussian measure $d\mu_0(A)$. This
 34 requirement has been one of the main obstructions to the above question. Our aim is to
 35 review a representation of $\ln \det_{\text{ren}}$ [18] that can accommodate these potentials and provides
 36 non-perturbative estimates of its large-field variations. In this letter we show that its growth
 37 is slowed down due to the presence of quasi-zero modes embedded in the space of allowed
 38 potentials.

39 Since we are calculating in Euclidean space we regard the field strength F to be static,
 40 four-dimensional magnetic field.

41 A quasi-zero mode arises from a sequence of eigenvalues $\lambda_1, \lambda_2, \dots$ generated by solving for
 42 random magnetic field's bound states. These are required to calculate the effective action.
 43 A sample magnetic field can generate a subset of eigenvalues, $\{\lambda_n\}$, that behave as $|\lambda_n| \rightarrow 1$
 44 in the large-field limit.

45 An eigenstate corresponding to one of these eigenvalues will be called QZM. Each of these
 46 modes reduce the growth of the effective action substantially.

47 The special case of vanishing binding energy is a zero mode that results in a negative
 48 contribution to the effective action without the need of a large-field limit. We expect these
 49 modes are sparse-requiring high symmetry-since the only known example of an admissible

50 zero mode in the effective action of (1) requires an intricate $O(2) \times O(3)$ symmetry [18].
 51 We will now summarize known results relevant to the functional integral below.

52 In contrast, QZMs only require a deep, wide well in a random F . Randomness is the
 53 essential point here as there is apparently no limit to the number of QZMs. A random QZM's
 54 probable low symmetry-degeneracy-can be offset by the number of bound states supported
 55 by a random potential, thereby driving the effective action increasingly less positive for large
 56 field values. We now proceed to show this.

57 A. Potentials

58 These have already been discussed in [18] which dealt with the electrodynamic sector of
 59 the electroweak model. To alleviate any source of confusion here we outline the derivation
 60 of $\det_{\text{ren}}$ in QED alone to make this letter reasonably self-contained.

61
 62 The potentials required to define $\det_{\text{ren}}$ and to support the functional measure $d\mu_0(A)$
 63 in (1) are sufficient if (1) $A_\mu \in \mathcal{S}'(\mathbb{R}^4)$ with a $\frac{1}{|x|}$ falloff, (2) $A_\mu \in \bigcap_{r>4} L^r(\mathbb{R}^4)$, and (3)
 64 $F \in \bigcap_{r>2}^{<4} L^r(\mathbb{R}^4)$ assuming that the fermion mass $m \neq 0$ in $\det_{\text{ren}}$.
 65

66 The final requirement is to introduce an ultraviolet cutoff by smoothing the potentials
 67 $A_\mu \in \mathcal{S}'$ with the functions of rapid decrease f_Δ

$$A_\mu^\Delta(x) = \int d^4y f_\Delta(x-y) A_\mu(y), \quad (2)$$

68 so that $A_\mu^\Delta \in C^\infty$ and hence is infinitely differentiable. A_μ^Δ and $F_{\mu\nu}^\Delta = \partial_\mu A_\nu^\Delta - \partial_\nu A_\mu^\Delta$ also
 69 satisfy conditions (2) and (3) above. The potentials A_μ have covariance

$$\int d\mu_0(A) A_\mu(x) A_\nu(y) = D_{\mu\nu}(x-y), \quad (3)$$

70 where $D_{\mu\nu}$ is the free photon propagator in a gauge fixed measure. Therefore the potentials
 71 A_μ^Δ have covariance

$$\int d\mu_0(A) A_\mu^\Delta(x) A_\nu^\Delta(y) = D_{\mu\nu}^\Delta(x-y), \quad (4)$$

72 where, $D_{\mu\nu}^\Delta$ has Fourier transform $\hat{D}_{\mu\nu} |\hat{f}_\Delta(k)|^2$, with $\hat{f}_\Delta(k) = 1$, $k^2 \leq \Delta^2$, $\hat{f}_\Delta(k) = 0$,
 73 $k^2 > \Delta^2$. Thus the smoothing process also plays the dual role of an ultraviolet regulator Δ .

In what follows all of the rough potentials A_μ in $\det_{\text{ren}}$ are replaced with the smooth potentials A_μ^Δ except in the measure $d\mu_0(A)$. From here on we will drop the superscript Δ ; it will only reappear again through its covariance (4), that is, when (1) is functionally integrated. We comment below on removing Δ .

We now state one of the most important results of this letter, namely that the operator $\Delta_A^{\frac{1}{2}} \sigma F \Delta_A^{\frac{1}{2}} \in \mathcal{S}_3$, [2], [7], [18] where

$$\Delta_A^{\frac{1}{2}}(x, y) \equiv \frac{1}{\pi} \int_0^\infty \frac{da}{\sqrt{a}} \left\langle x \left| \left[(D - A)^2 + m^2 + a \right]^{-1} \right| y \right\rangle, \quad \sigma_{\mu\nu} = [\gamma_\mu, \gamma_\nu] / 2, \quad \gamma_\mu^\dagger = -\gamma_\mu. \quad (5)$$

That is this operator belongs to the trace ideal $\mathcal{S}_3$ means that it is compact with discrete eigenvalues having finite multiplicity.

The integral (1) will give a non-perturbative result by depending only on D^Δ since its measure is Gaussian. This should simplify the limit $\Delta \rightarrow \infty$ of the result. Of course it is assumed that the integral (1) converges to which we now return.

B. $\det_{\text{ren}}$

We begin by recalling Schwinger's proper-time definition of $\ln \det_{\text{ren}}$ written as

$$\ln \det_{\text{ren}} (1 - \underline{e} S \mathcal{A}) = \frac{1}{2} \int_0^\infty \frac{dt}{t} \left(\text{Tr} \left\{ e^{-P^2 t} - \exp \left[- \left(D^2 + \frac{\underline{e}}{2} \sigma_{\mu\nu} F_{\mu\nu} \right) t \right] \right\} + \frac{\underline{e}^2 \|F\|^2}{24\pi^2} \right) e^{-tm^2}, \quad (6)$$

where $D_\mu = P_\mu - \underline{e} A_\mu$, $\|F\|^2 = \int d^4 x F^2$; $\underline{e}$, m are unrenormalized; ultraviolet divergences are suppressed by the regulator Δ introduced above.

From here on we omit $\underline{e}$ as it always multiplies A_μ and can be reinstated as required. The last term in (6) is a charge renormalization subtraction in the one-particle irreducible photon self-energy at zero momentum transfer following Eq. (C7), Appendix C [18]. It is extremely useful to rearrange (6) as

$$\begin{aligned} \ln \det_{\text{ren}} = \frac{1}{2} \int_0^\infty \frac{dt}{t} e^{-tm^2} & \left[4 \text{Tr} \left(e^{-P^2 t} - e^{-D^2 t} \right) - \frac{\underline{e}^2 \|F\|^2}{48\pi^2} \right. \\ & \left. + \text{Tr} \left(e^{-D^2 t} - \exp \left[- \left(D^2 + \frac{\underline{e}}{2} \sigma_{\mu\nu} F_{\mu\nu} \right) t \right] \right) + \frac{\underline{e}^2 \|F\|^2}{16\pi^2} \right], \end{aligned} \quad (7)$$

where the trace over spin was made in the first term to give a factor of 4. A volume cutoff on $\|F\|^2$ will be discussed below.

95 Then (7) becomes

$$\ln \det_{\text{ren}} = 2 \ln \det_{\text{SQED}} + \frac{1}{2} \int_0^\infty \frac{dt}{t} e^{-tm^2} \left[\text{Tr} \left(e^{-D^2 t} - \exp \left[- \left(D^2 + \frac{e}{2} \sigma_{\mu\nu} F_{\mu\nu} \right) t \right] \right) + \frac{e^2 \|F\|^2}{16\pi^2} \right] \quad (8)$$

96 . Here $\ln \det_{\text{SQED}}$ is the proper-time definition of the formal scalar QED determinant
 97 $\ln \left\{ \frac{[(P-A)^2 + m^2]}{(P^2 + m^2)} \right\}$ with on-shell charge renormalization

$$\ln \det_{\text{SQED}} = \int_0^\infty \frac{dt}{t} \left[\text{Tr} \left(e^{-P^2 t} - e^{-D^2 t} \right) - \frac{e^2 \|F\|^2}{192\pi^2} \right] e^{-tm^2}. \quad (9)$$

98 Care is needed to make sense of the remaining trace in (8) subject to the requirements (2)
 99 and (3) on the potentials listed above. This is done by breaking it up into two well-defined
 100 terms. Introduce the operator identity

$$e^{-t(D^2 + \frac{1}{2} e \sigma F)} - e^{-tD^2} = \int_0^t ds e^{-(t-s)(D^2 + \frac{1}{2} e \sigma F)} \frac{1}{2} e \sigma F e^{-sD^2}, \quad (10)$$

101 and iterate it twice to get terms of order σF and $(\sigma F)^2$ plus the remainder term. Insert
 102 these in (8). The σF term cancels in (8) due to the spin trace. The remainder term in (8)
 103 defines the determinant $\det_3$ as

$$\begin{aligned} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} e \sigma F \Delta_A^{\frac{1}{2}} \right) &= \int_0^\infty \frac{dt}{t} e^{-tm^2} \text{Tr} \left[\int_0^t ds_1 \int_0^{t-s_1} ds_2 \int_0^{t-s_1-s_2} ds_3 \right. \\ &\quad \times e^{-(t-s_1-s_2-s_3)(D^2 + \frac{1}{2} e \sigma F)} \frac{1}{2} e \sigma F e^{-s_3 D^2} \frac{1}{2} e \sigma F e^{-s_2 D^2} \frac{1}{2} e \sigma F e^{-s_1 D^2} \left. \right], \end{aligned} \quad (11)$$

104 with $\Delta_A^{\frac{1}{2}}$ defined in (5). This introduces the regularized determinant

$$\det_3 (1 + T) = \det \left[(1 + T) \exp \left(-T + \frac{1}{2} T^2 \right) \right], \quad (12)$$

105 where $T \in \mathcal{S}_3$. Its trace converges and depends on T^3 and higher powers of T . The equiva-
 106 lence of the two sides of (11) follows from Theorem 7.2 in [6]. Because of its inaccessibility
 107 and printing errors we rederive in Appendix B of [18]. Combining (9), (11) and the second
 108 iterate of (10) results in

$$\begin{aligned} \ln \det_{\text{ren}} &= 2 \ln \det_{\text{SQED}} + \frac{1}{2} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} e \sigma F \Delta_A^{\frac{1}{2}} \right) \\ &\quad + \frac{e^2}{8} \int_0^\infty \frac{dt}{t} e^{-tm^2} \left(\frac{1}{4\pi^2} \|F\|^2 - \text{Tr} \int_0^t ds_1 \int_0^{t-s_1} ds_2 e^{-(t-s_1-s_2)D^2} \sigma F e^{-s_2 D^2} \sigma F e^{-s_1 D^2} \right). \end{aligned} \quad (13)$$

109 The last term simplifies following Appendix C of [18] to give the three-term representation
 110 of $\det_{\text{ren}}$.

$$\begin{aligned} \ln \det_{\text{ren}} &= 2 \ln \det_{\text{SQED}} + \frac{1}{2} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} \underline{e} \sigma F \Delta_A^{\frac{1}{2}} \right) \\ &\quad + \underline{e}^2 \int_0^\infty dt e^{-tm^2} \left[\frac{1}{32\pi^2 t} \|F\|^2 - \frac{1}{2} \text{Tr} \left(e^{-tD^2} F_{\mu\nu} F_{\mu\nu} \Delta_A F_{\mu\nu} \right) \right]. \end{aligned} \quad (14)$$

111 Since the operator in $\det_3$ belongs to $\mathcal{S}_3$ we can use the operator identity

$$\begin{aligned} \ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} \underline{e} \sigma F \Delta_A^{\frac{1}{2}} \right) &= \ln \det \left[(1 + T) \exp \left(-T + \frac{1}{2} T^2 \right) \right] \\ &= \text{Tr} \left[\ln (1 + T) - T + \frac{1}{2} T^2 \right] \\ &= \sum_{n=1}^\infty \left[\ln (1 + \lambda_n) - \lambda_n + \frac{1}{2} \lambda_n^2 \right]. \end{aligned} \quad (15)$$

112 The eigenvalues $\{\lambda_n\}$ of $\Delta_A^{\frac{1}{2}} \sigma F \Delta_A^{\frac{1}{2}}$ occur in pairs $(-\lambda, \lambda)$ following C -invariance (Appendix
 113 D, [18]). We can alternatively replace $\ln \det_3$ in [14] with an eigenvalue sum

$$\ln \det_3 \left(1 + \Delta_A^{\frac{1}{2}} \frac{1}{2} \underline{e} \sigma F \Delta_A^{\frac{1}{2}} \right) = \sum_{n=1}^\infty [\ln (1 - \lambda_n^2) + \lambda_n^2], \quad (16)$$

114 where the sum is only over positive eigenvalues. Since $\ln \det_3$ is real and finite then $0 <$
 115 $\lambda_n < 1$ for all n so that $\ln \det_3 < 0$. The sum in (16) has an infinite radius of convergence.
 116 That $\ln \det_3 < 0$ is due to the paramagnetism of a charged spin- $\frac{1}{2}$ fermion in a static, four
 117 dimensional magnetic field.

118 On average the eigenvalues are lowered by $F_{\mu\nu}$ following the Peierls-Bogoliubov inequality
 119 derived from Klein's inequality [19–21]. It is already apparent in (15) and (16) that the
 120 lowest-lying eigenvalues will dominate the convergence of (1). In the large magnetic field
 121 limit due to QZMs defined above. That these were overlooked since 1951 is probably because
 122 they are essentially non-perturbative.

123 II. ANALYTIC CALCULATION OF A QZM.

124 The eigenvalues of (16) are obtained from

$$\frac{\underline{e}}{2} \Delta_A^{\frac{1}{2}} \sigma F \Delta_A^{\frac{1}{2}} \varphi_n = -\lambda_n \varphi_n, \quad (17)$$

125 where $\varphi \in L^2$. Let $\Delta_A^{\frac{1}{2}}\varphi = \psi_n$ and obtain

$$\left[(P - \underline{e}A)^2 + \frac{\underline{e}}{2\lambda_n}\sigma F \right] \psi_n = -m^2\psi_n, \quad (18)$$

126 where $\psi_n \in L^2$ as shown at the end of Appendix A of [18]. From here on we assume $\underline{e}$ is
 127 real. The $\{\lambda_n\}$ assume the role of coupling constants whose discrete values result in bound
 128 states with energy m^2 . The states ψ_n have definite chirality since γ_5 commutes with σ by
 129 choosing a representation of γ_5 with diagonal elements $\pm \mathbb{1}_2$.

130 Therefore we need only deal with 2D chiral eigenstates $\psi_n^\dagger$. To demonstrate an analytically
 131 solvable QZM we choose an $O(2) \times O(3)$ symmetric potential. These have the standard
 132 form [22–24].

$$A_\mu(x) = M_{\mu\nu}x_\nu a(r), \quad (19)$$

133 here $M_{\mu\nu}$ is the antiself-dual, antisymmetric matrix with nonvanishing elements $M_{12} =$
 134 $M_{30} = 1$ and $r^2 = x_\mu^2$.

135 This potential is chosen solely to simplify the analysis; QZMs have no particular symmetry.
 136 The eigenstates of (18) are represented by a 1×4 column matrix [24] whose first two rows
 137 consist of the positive chirality states

$$\psi_n = r^{-2j-\frac{3}{2}} \begin{bmatrix} \mathcal{D}_{M-\frac{1}{2},m}^j \rho_1(r) \\ \mathcal{D}_{M+\frac{1}{2},m}^j(x) \rho_2(r) \end{bmatrix}, \quad (20)$$

138 where $\mathcal{D}_{m_1,m_2}^j$ are the four dimensional rotation matrices [24–26] and $2j = 0, 1, \dots$; $-j \leq$
 139 $m \leq j$.

140 Inserting (20) into (18) gives

$$\left[\frac{d^2}{dr^2} + \frac{(2j+1)^2 - \frac{1}{4}}{r^2} + (4M \mp 2)\underline{e}a + \underline{e}^2 r^2 a^2 \pm \frac{\underline{e}}{\lambda_n^+} \left(4a + r \frac{da}{dr} \right) \right] \rho_{1,2} = -m^2 \rho_{1,2}, \quad (21)$$

141 where the upper (lower) sign applies to ρ_1 (ρ_2) and λ_n^+ denotes a positive chirality eigenvalue
 142 that results in L^2 bound states. In fact it is already evident from (18) that only $0 < \lambda_n < 1$
 143 results in a bound state when $\underline{e}$ is real.

144 We chose one of the simplest potentials (19) with

$$a(r) = \begin{cases} \frac{C}{R^2} \left(\frac{r}{R} \right)^{\varepsilon-2} + \frac{(2-\varepsilon)C-2\nu}{R^3} r + \frac{(\varepsilon-3)C+3\nu}{R^2} & \text{if } r \leq R \\ \frac{\nu}{r^2}, & r > R \end{cases} \quad (22)$$

giving $F_{\mu\nu}^2 = 8r^2a'^2 + 16a^2 + 16r^2aa'$, $a' = \frac{da}{dr}$, and $\varepsilon > 2$ so that $F \in \bigcap_{p>2} L^p$. The constant C can be positive or negative, and $\nu > 0$. In order that $A_\mu \in C^\infty$ we assume that there is an infinitesimal zone of smooth curvature centered at $r = R$. This has no effect on the present analysis.

Setting $M = -j - \frac{1}{2}$ to ensure a bound state in (21) and letting $\rho_2 = f$ with $r \leq R$, $\underline{e} \gg 1$.

$$\left[\frac{d^2}{dr^2} + \frac{\frac{1}{4} - (2j+1)^2}{r^2} + 4 \left(j + \frac{1}{\lambda} \right) \underline{e}a - \underline{e}^2 r^2 a^2 + \frac{\underline{e}}{\lambda} r \frac{da}{dr} \right] = m^2 f. \quad (23)$$

For $r > R$ let $f = r^{\frac{1}{2}}g$ so that (23) becomes

$$g'' + \frac{1}{r}g' - \left(m^2 + \frac{(2j+1-\underline{e}\nu)^2 + 2(1-\frac{1}{\lambda})\underline{e}\nu}{r^2} \right) g = 0. \quad (24)$$

This has the decaying solution $K_\alpha(mr)$ which is a modified Bessel function with $\alpha = [(2j+1-\underline{e}\nu)^2 + 2(1-\frac{1}{\lambda})\underline{e}\nu]^{\frac{1}{2}}$. The eigenvalue λ is fixed by the boundary condition at $r = R$:

$$\frac{Rf'(R)}{f(R)} = \frac{1}{2} + \frac{RK'_\alpha(mR)}{K_\alpha(mR)}. \quad (25)$$

The analysis is simplified by setting $mR \ll 1$ and making the small mass expansions $F = Bf_0(1 + m^2f_2 + \mathcal{O}(m^4f_4))$, $\lambda = 1 - m^2\delta_2 + \mathcal{O}(m^4\delta_4)$, B is a normalization constant. Here f_0 is the solution of (23) when $m = 0$ and $\lambda = 1$ and $0 \leq r \leq R$ where $f_0 = r^{2j+\frac{3}{2}}e^{-\underline{e}\int_0^r ds sa(s)}$. The two sides of (25) can be matched in powers of m to obtain for $\underline{e}\nu \gg 1$

$$\lambda = 1 - \frac{m^2R^2}{2\underline{e}\nu} \left(1 + \mathcal{O}\left(\frac{1}{\underline{e}\nu}\right) \right), \quad (26)$$

following Appendix E of [18]. Such an eigenvalue approaching 1 from below is the signature of a QZM as referred to in Sec. I above.

We conclude that for this mode its contribution to $\ln \det_3$ in (16) is

$$\ln \det_3 = -\frac{\underline{e}^2\nu^2}{2} \ln \left(\frac{\underline{e}\nu}{m^2R^2} \right) \left(1 + \mathcal{O}\left(\frac{1}{\underline{e}\nu}\right) \right) + \sum_{n=2}^{\infty} [\ln(1 - \lambda_n^2) + \lambda_n^2]. \quad (27)$$

All remaining bound states from (19), (22) contribute a negative term to (27). Presumably (26) is the lowest lying bound state eigenvalue. There might remain a band of closely spaced bound states for $n \geq 2$, each with degeneracy of $\mathcal{O}(\underline{e}^2\nu^2)$.

All of the bound states of (17) contribute a negative term to $\ln \det_3$ irrespective of their symmetry or lack thereof.

167 **III. COMPLETION OF $\det_{\text{ren}}$**

168 The first and third terms in (14) have already been calculated in the large limit in (1).
 169 From Eq. (A24) in [23]. We found

$$\ln \det_{\text{SQED}} \stackrel{\underline{e}\mathcal{F} \rightarrow \infty}{\sim} -\frac{\underline{e}^2 \|F\|^2}{192\pi^2} \ln \left(\frac{\underline{e}\mathcal{F}}{m^2} \right) + \mathcal{O}(\underline{e}\mathcal{F}), \quad (28)$$

170 where the amplitude of $F_{\mu\nu}$ is represented by the parameter $\mathcal{F}$ with dimension L^{-2} . The
 171 third term is referred to here as the charge renormalization term, denoted by Π in Eq.(6.43)
 172 in [18] yielding

$$\Pi \stackrel{\underline{e}\mathcal{F} \rightarrow \infty}{\sim} -\frac{\underline{e}^2 \|F\|^2}{32\pi^2} \ln \left(\frac{\underline{e}\mathcal{F}}{m^2} \right) + R_1, \quad (29)$$

173 with $\lim_{\underline{e}\mathcal{F} \rightarrow \infty} \frac{R_1}{[(\underline{e}\mathcal{F})^2 \ln(\underline{e}\mathcal{F})]} = 0$.

174 In order to compare (28) and (29) with $\ln \det_3$ in (21) note that $\max F_{\mu\nu}^2 = \frac{8\nu^2}{R^4} \equiv \mathcal{F}^2$ for
 175 $0 \leq R < \infty$.

176 Accordingly, we set $\underline{e}\mathcal{F} = \frac{\underline{e}\nu}{R^2}$ for $\underline{e}\nu \gg 1$. Then from (14), (27), (28) and (29)

$$\begin{aligned} \ln \det_{\text{ren}} \stackrel{\underline{e}\mathcal{F} \rightarrow \infty}{\sim} & \frac{\underline{e}^2 \|F\|^2}{32\pi^2} \ln \left(\frac{\underline{e}\mathcal{F}}{m^2} \right) + R_1 - \frac{\underline{e}^2 \|F\|^2}{192\pi^2} \ln \left(\frac{\underline{e}\mathcal{F}}{m^2} \right) + \mathcal{O}(\underline{e}\mathcal{F}) \\ & - \frac{\underline{e}^2 \mathcal{F}^2 R^4}{2} \ln \left(\frac{\underline{e}\mathcal{F}}{m^2} \right) \left[1 + \mathcal{O} \left(\frac{1}{\underline{e}\nu} \right) \right] + \sum_{n=2}^{\infty} [\ln(1 - \lambda_n^2) + \lambda_n^2]. \end{aligned} \quad (30)$$

177 Additional zero mode contributions to $\ln \det_{\text{ren}}$ such as (6.45), (6.46) are omitted as they
 178 are not random variables and therefore not relevant here.

179

180 **IV. $\det_3$: GENERAL PROPERTIES**

181 (a) $\det_3(1 + \underline{e}T)$, $T = \Delta_{A^2}^{\frac{1}{2}} \sigma F \Delta_A^{\frac{1}{2}}$ is well defined by a series for $|\underline{e}|$ small and defined by
 182 analytic continuation to an entire function of $\underline{e}$ considered as a complex variable, which
 183 means that it is holomorphic in the entire complex plane.

184 (b) Let $\lambda_1(T), \dots$ be the eigenvalues of $T \in \mathcal{S}_3$. Then $\sum_{n=1}^{\infty} |\lambda_n|^{-3-\varepsilon} < \infty$, $\varepsilon > 0$;

185 (c) By Th. 6.2 of [9],

$$\det_3(1 + \underline{e}T) = \prod_{m=1}^{\infty} \left[(1 - \underline{e}\lambda_m(T)) \exp \left(\underline{e}\lambda_m(T) + \frac{(\underline{e}\lambda_m(T))^2}{2} \right) \right]. \quad (31)$$

The reality of $\det_3$ for real $\underline{e}$ together with C -invariance require that the eigenvalues appear as real pairs $\lambda_n, -\lambda_n$ or as quartets $\pm\lambda_n, \pm\bar{\lambda}_n$ for $m \neq 0$.

(d) $\det_3$ removes the $\mathcal{O}(\underline{e})$ term. The remaining $\mathcal{O}(\underline{e}^2)$ term is replaced with the on-shell renormalized term Π whose asymptotic result is (6.43) in [18]. This result was combined with $\ln \det_{\text{SQED}}$ in (28)

(e) The growth of $\det_3$ for $|\underline{e}|\mathcal{F} \rightarrow \infty$ depends on the distribution of its zeros $-\frac{1}{\lambda_1}, \dots$ [27].

V. CONCLUSION

The opening question of this letter remains unanswered. Progress has been made here by narrowing it to Eq. (30). Its first two terms sum to a positive value. The last term, $\ln \det_3$, is always negative, and hence is the controlling determinant for asymptotic freedom in QED. Its factor, $\ln\left(\frac{\underline{e}\mathcal{F}}{m^2}\right)$, is always present due to QZMs. Furthermore, it does not require a cutoff. What we do not know is the overall falloff of $\ln \det_3$. What we do know is that the distribution of the eigenvalues of the compact operator $T \in \mathcal{S}_3$ in the complex $\underline{e}$ -plane determines the growth of $|\det_3|$ for $|\underline{e}\mathcal{F}| \rightarrow \infty$ [27]. Knowing this should be enough to settle the question of asymptotic freedom in QED.

A convergent result for a QED process (1) will contain an ultraviolet cutoff Δ in the free photon propagator following the Gaussian integration (3). Recall that it was introduced to smooth the tempered distributions supporting the functional integral. A volume cutoff also remains. The ability to remove these cutoffs by some non-perturbative renormalization procedure leads directly to the long-standing question of whether QED is an interacting theory.

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